

Mathematical methods list of oral exam questions

1. Fourier sine and cosine series: definitions, calculation of Fourier coefficients, approximation by Fourier series, Parseval's identity.
2. Fourier sine and cosine transform.
3. Complex Fourier transform.
4. Fourier Transform identities: shift, rescaling, derivatives and integrals.
5. Dirac delta.
6. Discrete and Fast Fourier Transforms.
7. First order ordinary differential equations: geometric meaning, direction fields, Euler's Method.
8. Separable ODEs.
9. First order linear ODEs. Solution of nonhomogeneous equation. Existence and uniqueness of solution for initial value problem.
10. Homogeneous Linear ODEs of Second Order.
11. Homogeneous Linear ODEs with Constant Coefficients.
12. Nonhomogeneous linear ODEs with Constant Coefficients
13. Basic Theory of Systems of ODEs. Wronskian
14. Constant-Coefficient Systems. Phase Plane Method. Criteria for Critical Points. Stability
15. Vibrating String, Wave Equation, Solution by Separating Variables. Use of Fourier Series
16. D'Alembert's Solution of the Wave Equation.
17. Heat Equation, Solution by Separating Variables. Use of Fourier Series
18. Heat Equation in infinite space. Use of Fourier transform, solution by convolution, heat kernel.
19. Laplace's equation, Solution by Separating Variables. Use of Fourier Series
20. Complex numbers, complex algebra, polar form. Complex Derivative. Analytic Function, Cauchy-Riemann Equations. Laplace's Equation
21. Complex integral, Cauchy's integral formula.